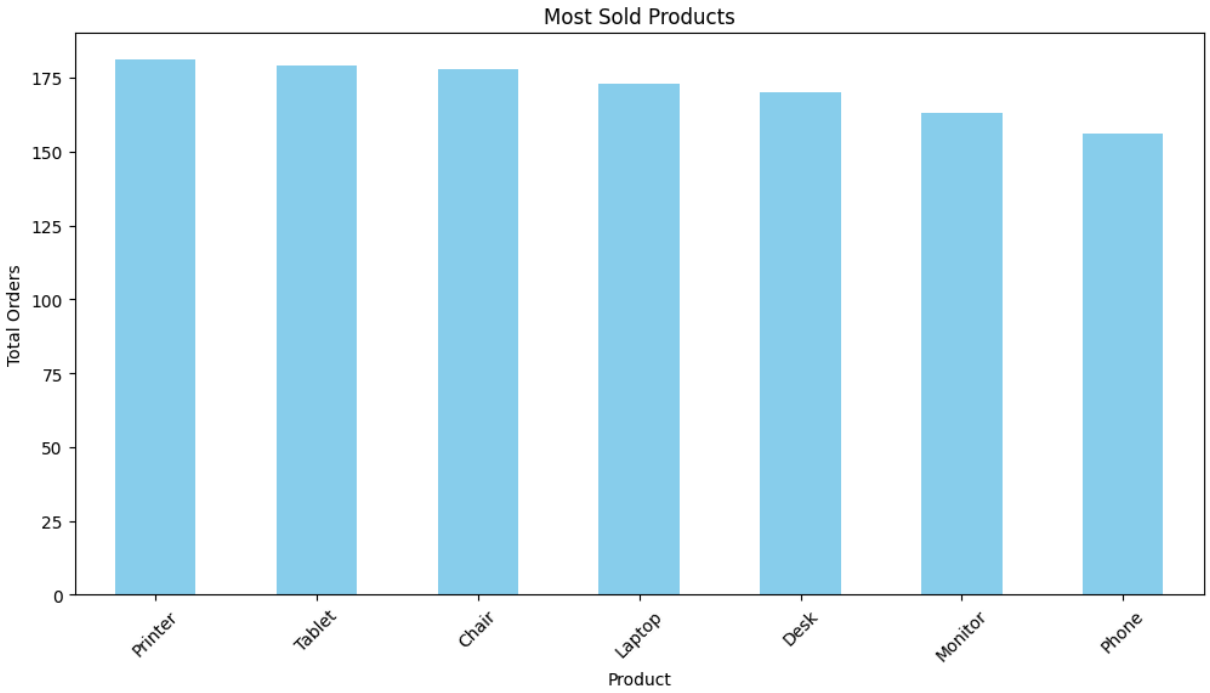
Chart 1: Most Sold Products

In [12]:
```
import pandas as pd
df = pd.read_excel('data.xlsx')
```

In [13]:
```
import matplotlib.pyplot as plt
plt.figure(figsize=(12,6))
df['Product'].value_counts().plot(kind='bar', color='skyblue')
plt.title('Most Sold Products')
plt.xlabel('Product')
plt.ylabel('Total Orders')
plt.xticks(rotation=45)
plt.show()
```

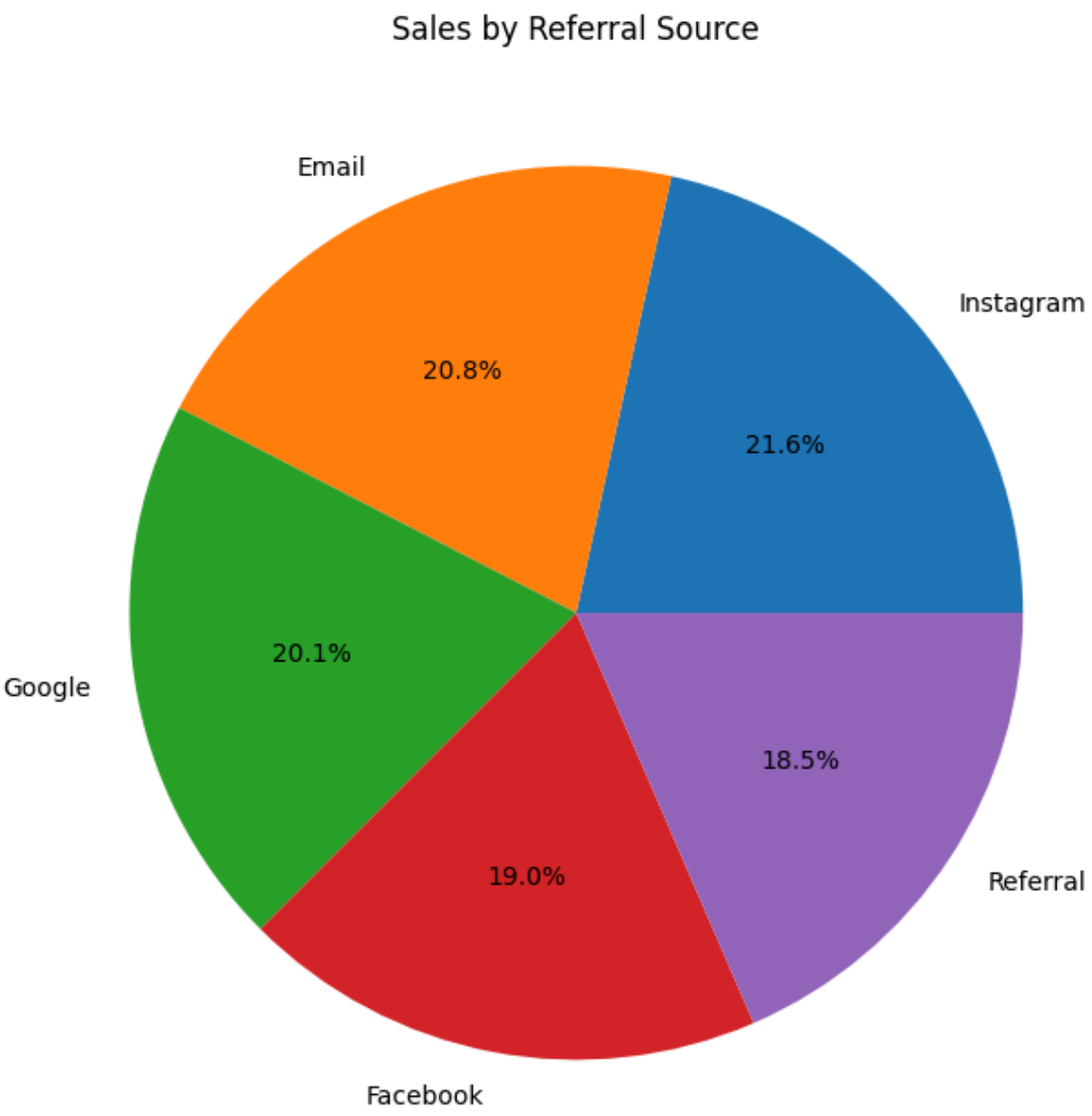


Insight 1: Most Sold Products

Printer (181), Tablet (179) and Chair (178) are the most ordered products, while Phone is the least at 156. Order volume is fairly evenly spread, so inventory should be prioritized for the top three.

Chart 2: Sales by Referral Source

In [14]:
```
plt.figure(figsize=(8,8))
df['ReferralSource'].value_counts().plot(kind='pie', autopct='%1.1f%%')
plt.title('Sales by Referral Source')
plt.ylabel('') # ye line title ke neeche 0 hatane ke liye
plt.show()
```

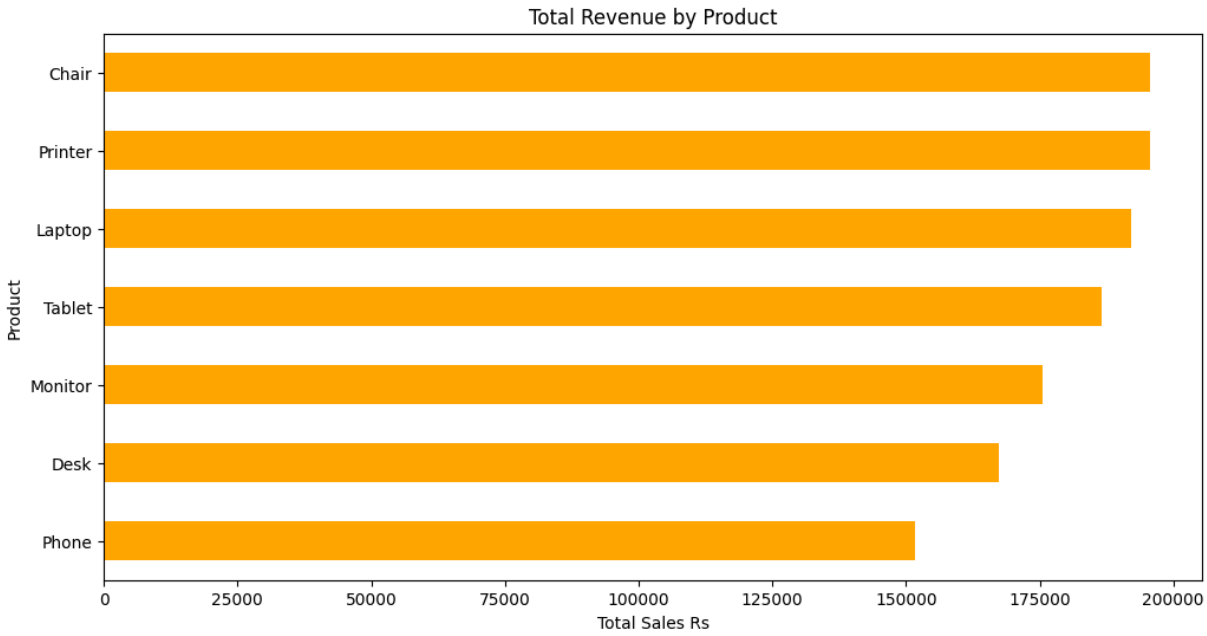


Insight 2: Sales by Referral Source

Instagram is the top referral source at 21.6%, followed by Email at 20.8% and Google at 20.1%. Facebook (19.0%) and Referral (18.5%) are lowest. Differences are small, so marketing spend can be tested on Instagram and Email first.

Chart 3: Total revenue by Products

In [15]:
```
plt.figure(figsize=(12,6))
df.groupby('Product')['TotalPrice'].sum().sort_values().plot(kind='barh', color='or
plt.title('Total Revenue by Product')
plt.xlabel('Total Sales Rs')
plt.ylabel('Product')
plt.show()
```

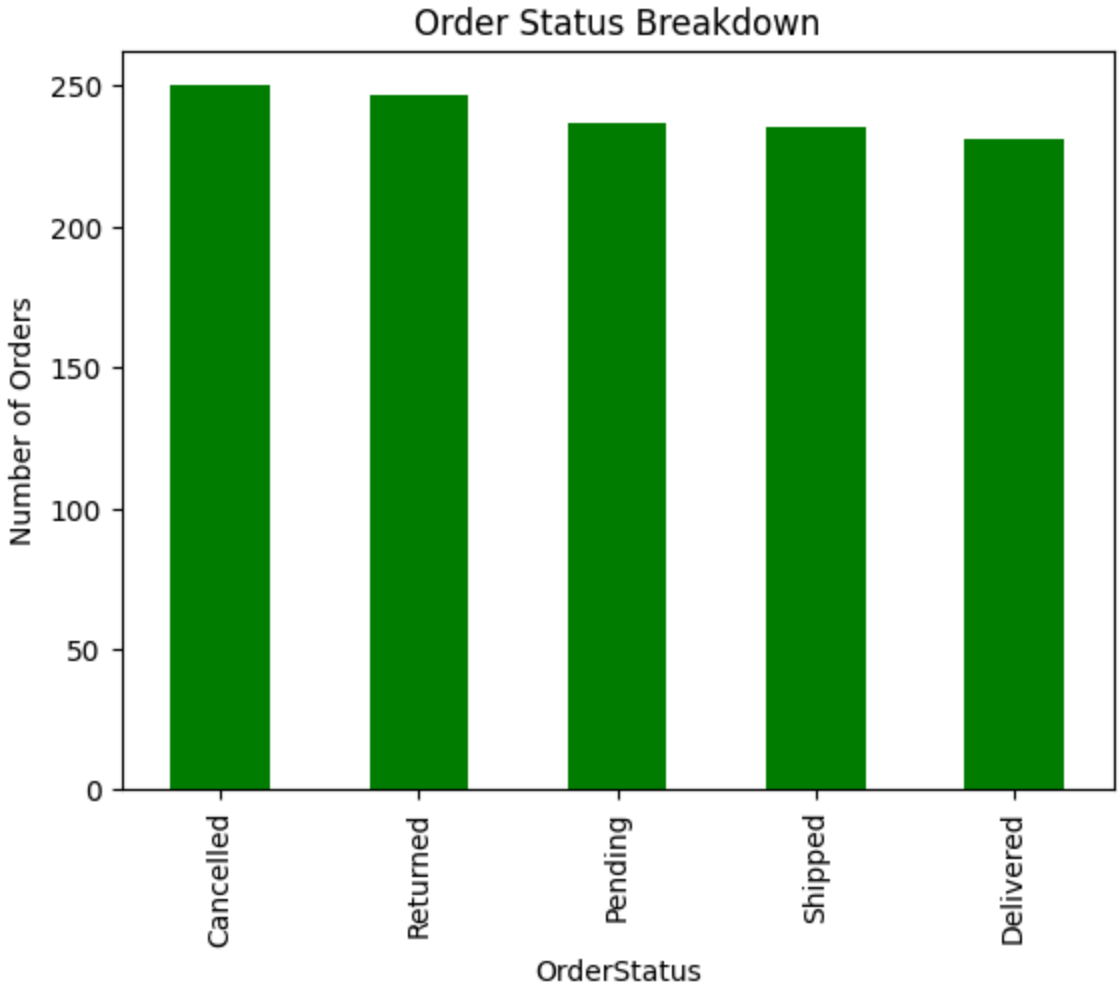


Insight 3: Total Revenue by Product

Chair and Printer generate the highest revenue (~Rs 196K each), while Phone generates the lowest (~Rs 152K).

Chart 4: Order status Breakdown

In [16]:
```
df['OrderStatus'].value_counts().plot(kind='bar', color='green')
plt.title('Order Status Breakdown')
plt.ylabel('Number of Orders')
plt.show()
```



Insight 4: Order Status Breakdown [Most Important]

Cancelled (250) and Returned (247) orders each exceed Delivered (231). Together they make up 41.4% of all 1,200 orders, a fulfillment or product-quality issue worth investigating.